



ADVENTIST UNIVERSITY OF CENTRAL AFRICA

Exercise: Employment System Design 2024

Course: Database Development with PL/SQL INSY 8311

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Stored Procedure Exercises

Question 1: VAT Calculation Procedure for RRA

The Rwanda Revenue Authority (RRA) maintains a database with customer transaction data. Your task is to create a stored procedure that calculates Value Added Tax (VAT) for each transaction. The VAT rate is 18%, which should be stored as a constant value.

Requirements:

1. Define a global variable to store the VAT rate (18%).
2. Use a local variable inside the procedure to store intermediate calculations.
3. Accept the total amount of a transaction as a parameter.
4. Compute the VAT by applying the 18% rate and return both the VAT amount and the total amount after VAT.
5. The procedure should return the calculated VAT and the final amount (total amount + VAT).

Procedure Implementation:

Write a stored procedure named `Calculate_VAT` that follows the above requirements.

Procedure Call Example:

Once you have created the procedure, call it using a transaction amount of 100,000 RWF and display the VAT and final amount.

Question 2: Employee Bonus Calculation

A company wants to calculate bonuses for its employees based on their salary and performance rating. The bonus is calculated as follows:

- If the performance rating is 'A', the bonus is 20% of the salary.
- If the performance rating is 'B', the bonus is 10% of the salary.
- If the performance rating is 'C', the bonus is 5% of the salary.

Requirements:

1. Create a stored procedure named `Calculate_Bonus` that takes an employee ID, salary, and performance rating as input.
2. Calculate the bonus based on the performance rating.
3. Return the calculated bonus.

Procedure Call Example:

Call the procedure for an employee with a salary of 50,000 RWF and a performance rating of 'A'. Display the bonus amount.

Question 3: Customer Discount Calculation

An e-commerce platform wants to calculate discounts for customers based on their total purchase amount. The discount rules are as follows:

- If the purchase amount is greater than 100,000 RWF, apply a 10% discount.
- If the purchase amount is between 50,000 RWF and 100,000 RWF, apply a 5% discount.
- If the purchase amount is less than 50,000 RWF, no discount is applied.

Requirements:

1. Create a stored procedure named `Calculate_Discount` that takes the purchase amount as input.
2. Calculate the discount based on the rules above.
3. Return the discount amount and the final amount after applying the discount.

Procedure Call Example:

Call the procedure for a purchase amount of 120,000 RWF and display the discount and final amount.

Question 4: Student Grade Calculation

A school wants to calculate the final grade for students based on their marks in three subjects: Math, Science, and English. The final grade is calculated as the average of the three marks.

Requirements:

1. Create a stored procedure named `Calculate_Grade` that takes the marks for Math, Science, and English as input.
2. Calculate the average of the three marks.
3. Return the final grade.

Procedure Call Example:

Call the procedure for a student with marks 85, 90, and 78 in Math, Science, and English, respectively. Display the final grade.

Question 5: Product Price Update

A retail store wants to update the price of a product by applying a percentage increase. The store manager wants to automate this process using a stored procedure.

Requirements:

1. Create a stored procedure named `Update_Product_Price` that takes the product ID and the percentage increase as input.
2. Update the product's price in the products table by applying the percentage increase.
3. Return the updated price.

Procedure Call Example:

Call the procedure for a product with ID 101 and a 15% price increase. Display the updated price.

Question 6: Employee Salary Increment

A company wants to increment the salary of employees based on their years of service. The increment rules are as follows:

- If the years of service are greater than 10, increment the salary by 15%.
- If the years of service are between 5 and 10, increment the salary by 10%.
- If the years of service are less than 5, increment the salary by 5%.

Requirements:

1. Create a stored procedure named Increment_Salary that takes the employee ID and years of service as input.
2. Calculate the salary increment based on the rules above.
3. Update the employee's salary in the employees table.

Procedure Call Example:

Call the procedure for an employee with ID 202 and 8 years of service. Display the updated salary.

Question 7: Order Total Calculation

An online store wants to calculate the total cost of an order, including shipping fees. The shipping fee is 5,000 RWF for orders below 50,000 RWF and free for orders above 50,000 RWF.

Requirements:

1. Create a stored procedure named Calculate_Order_Total that takes the order amount as input.
2. Calculate the shipping fee based on the order amount.
3. Return the total cost (order amount + shipping fee).

Procedure Call Example:

Call the procedure for an order amount of 45,000 RWF and display the total cost.

Question 8: Loan Interest Calculation

A bank wants to calculate the interest on a loan based on the principal amount, interest rate, and loan duration (in years). The formula for simple interest is:

$$\text{Interest} = \text{Principal} \times \text{Rate} \times \text{Time}$$

Requirements:

1. Create a stored procedure named Calculate_Loan_Interest that takes the principal amount, interest rate, and loan duration as input.
2. Calculate the interest using the formula above.
3. Return the calculated interest.

Procedure Call Example:

Call the procedure for a principal amount of 1,000,000 RWF, an interest rate of 8%, and a loan duration of 5 years. Display the interest.

Question 9: Customer Age Verification

An online platform requires customers to be at least 18 years old to register. The platform wants to verify the age of customers using their date of birth.

Requirements:

1. Create a stored procedure named Verify_Age that takes the customer's date of birth as input.
2. Calculate the customer's age.
3. Return a message indicating whether the customer is eligible to register (age >= 18) or not.

Procedure Call Example:

Call the procedure for a customer born on January 1, 2005, and display the eligibility message.

Question 10: Employee Retirement Eligibility

A company wants to check if an employee is eligible for retirement based on their age. The retirement age is 60.

Requirements:

1. Create a stored procedure named `Check_Retirement_Eligibility` that takes the employee's date of birth as input.
2. Calculate the employee's age.
3. Return a message indicating whether the employee is eligible for retirement (age ≥ 60) or not.

Procedure Call Example:

Call the procedure for an employee born on January 1, 1960, and display the eligibility message.

Question 10: Employee Retirement Eligibility

A company wants to check if an employee is eligible for retirement based on their age. The retirement age is 60.

Requirements:

1. Create a stored procedure named `Check_Retirement_Eligibility` that takes the employee's date of birth as input.
2. Calculate the employee's age.
3. Return a message indicating whether the employee is eligible for retirement (age ≥ 60) or not.

Procedure Call Example:

Call the procedure for an employee born on January 1, 1960, and display the eligibility message.

Question 11: PL/SQL Question for Students:

You are tasked with writing a PL/SQL block to fetch and display employee details from the employees table. The employees table contains the following columns: `employee_id`, `first_name`, `last_name`, `salary`, and `hire_date`.

Instructions:

1. Declare a **record variable** using `%ROWTYPE` that will hold a single row of data from the employees table.
 2. Use a **SELECT INTO statement** to retrieve the details of an employee with `employee_id = 3`.
 3. Display the following details for the employee:
 - Employee ID
 - First Name
 - Last Name
 - Salary
 - Hire Date
 4. Ensure that the data is displayed using the `DBMS_OUTPUT.PUT_LINE` statement.
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Question 11: Anonymous PL/SQL Question: Insurance Company Scenario

An insurance company wants to check if a policyholder is eligible for a specific insurance benefit based on their age. The eligibility age for the insurance benefit is 60.

Instructions:

1. Write an **anonymous PL/SQL block** that calculates the **age of a policyholder** from their **date of birth**.
2. Use the policyholder's **date of birth** to determine if they are eligible for the insurance benefit. The eligibility condition is:
 - If the policyholder's age is **greater than or equal to 60**, they are eligible.
 - If the policyholder's age is **less than 60**, they are not eligible.

3. Output a message indicating whether the policyholder is eligible for the benefit or not.

Instructions for Your Anonymous PL/SQL Block:

- Declare a **date of birth** for the policyholder (for example: January 1, 1960).
- Calculate the **age** of the policyholder based on the **current date**.
- If the policyholder's age is **60 or older**, print: Eligible for insurance benefit.
- If the policyholder's age is **below 60**, print: Not eligible for insurance benefit.

"Whatever you do, work at it with all your heart, as working for the Lord, not for human masters."

— *Colossians 3:23 (NIV)*